LOURDES MATHA
COLLEGE OF SCIENCE & TECHNOLOGY
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Workbook

20MCA283-Deep Learning

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DEEP LEARNING (20MCA283)

MODULE-I

PART –A (3 marks each)

1. Describe sigmoid activation functions.

Sigmoid Activation function is widely used in backpropagation network because of relationship b/w the value of function at the point and the derivative during Computation.

There are two types of Sigmoidal function

i, Binary Sigmoidal function (Range 0-1)

$$f(x) = \frac{1}{1+e^{-\lambda x}}, \quad f'(x) = \lambda f(x)[1-f(x)],$$

λ = Steepness parameter,

ii, Bipolar Sigmoidal function (Range -1 to 1)

$$f(x) = \frac{2}{1+e^{-\lambda x}} - 1$$

$$f'(x) = \frac{\lambda}{2}[1+f(x)][1-f(x)]$$

2. Write the gradient descent algorithm.

Iteratively calculate the next step point using gradient of the current position, then scales it by a learning rate and subtract the obtained value from the current position.

parameters:

η - learning rate : Scales the gradient and thereby controls the step size. When η becomes smaller the longer gradient descent converges.

Algorithm :

Step 1: Choose a starting point

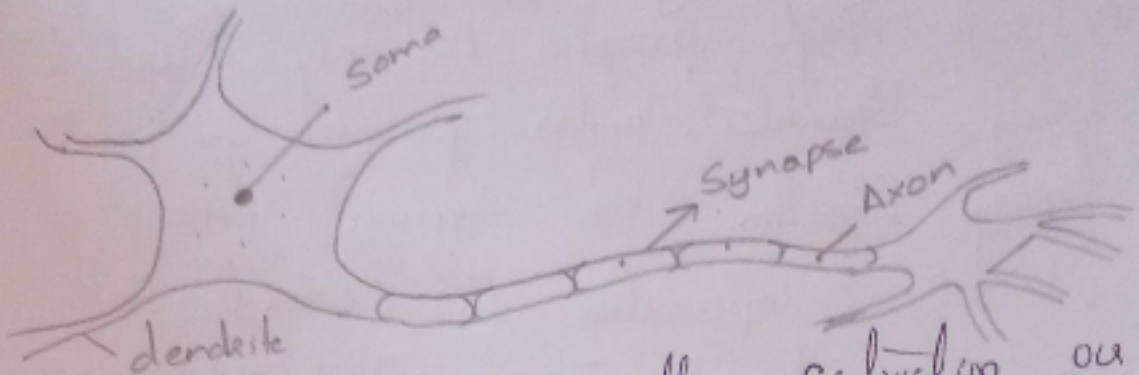
Step 2: Calculate gradient at this point

Step 3: Make a scaled step in the opposite direction to the gradient descent

Step 5: maximum no: of iteration is reached

Step 6: Step size is smaller than tolerance

3. Describe the model of a biological neuron.



1. The majority of neurons encode their activation or o/p as series of brief electrical pulses.

2. The neurons cell body (Soma) process the incoming activation and converts them into o/p activations.

3. The neuron's nucleus contains the genetic material in the form of DNA. This exists in most types of cells not just neurons.

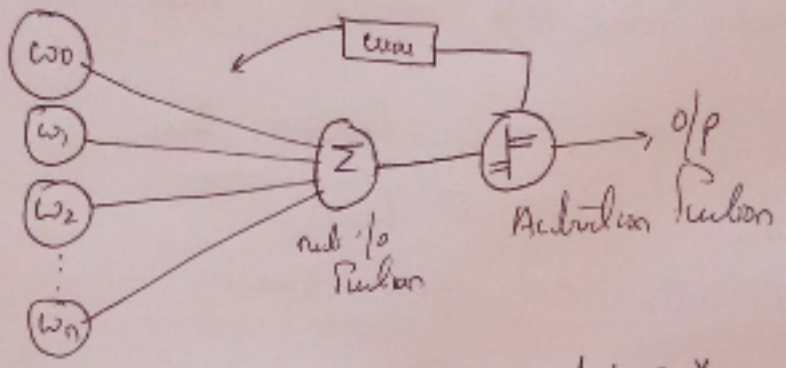
4. Dendrites are fibres which emanate from cell body and provide the receptive zone that receive activation from the neurons.

5. Axons are fibres acting as transmission line. activation to other neurons.

6. The junction that allows signal transmission b/w the axon end dendrites can called synapse. The process of transmission is by diffusion of chemical called neuro transmitters across the synaptic cells.

4. Explain Perceptron learning algorithm.

perceptron is a building block of an Artificial Neural network. Single perceptron can only be used to implement binary Separable function. perceptron is a machine learning algorithm for Supervised learning of binary classifiers. inperceptron the weight coefficient is automatically learned. The activation function applies a step rule to check whether the weighted function is greater than zero



Step 1: $\sum w_i * x_i = x_1 * w_1 + x_2 * w_2 \dots w_n * x_n$

$\sum w_i * x_i + b$

Step 2: $y = f(\sum w_i * x_i + b)$

Algorithm

P ← i/p with label 1;
N ← i/p with label 0;
initialize randomly
while ! Convergence do
pick random x ∈ P ∪ N;
if x ∈ P and wx < 0 then
w = w + x
end
if x ∈ N and wx ≥ 0 then
w = w - x;
end;
end;